PERSONAL SYSTEMS MEMO · V15 · LOCAL OWNER REVIEW

Pattern Recognition / *The Discrimination Layer*

Why repeated reports can still amount to one origin—and what an AI system should preserve before it generates.

WORKING PROPOSITION · CONCEPTUAL SYNTHESIS

Make the judgment before generation visible.

Some evidence-sensitive AI workflows may benefit from an explicit, inspectable responsibility for deciding what context can influence generation. Whether that visibility improves outcomes enough to justify its cost is the empirical question—not a conclusion.

TERM BOUNDARY

Discrimination means technical differentiation among information and possible actions—not social classification and not a model discriminator.

NO STUDY RUN · NO EMPIRICAL RESULTS · NOT PUBLISHED

The semantic local HTML reader is the canonical accessible surface. This PDF is an untagged visual/print companion.

Two tracks, one visible truth boundary

Read the essay for the argument; use the Lab to inspect the unrun protocol. The two surfaces meet at the same receipt and never blur concept with result.

ESSAY + EXPLORE

The conceptual track

- A concrete counting error and an exact fictional receipt.
- Six mechanism families and eleven inspectable responsibilities.
- Verified precedents, objections, retirement tests, and limits.

LAB

The empirical track

- One frozen-model supplied-cue question; F0/F1/F2 only.
- Fixed synthetic denominators, endpoints, stop gates, and null handling.
- Optional natural-syndication T1 stays descriptive and separate.

MAXIMUM CURRENT CLAIM

V15 may improve the conceptual synthesis and execution readiness. It does not establish provenance discovery, real-world independence, factual correctness, human benefit, field transfer, or a validated mechanism.

Suggested owner path

TIME	INSPECT	DECIDE
0–5 min	Opening, receipt, distinctions	Does the thesis survive the example?
5–12 min	Map, components, objections	Is the synthesis useful enough to keep?
12–20 min	Lab, T1, negative-result contract	Is the offline program ready for a later authorization choice?

01 · THE COUNTING ERROR

Nine reports can still amount to one origin.

Different headlines, layouts, and wording do not create new roots. The reports may remain useful observations, but recurrence cannot silently become corroboration.

THE FLAT SUMMARY SAYS

“Nine sources agree that the new tool is broadly validated.”

OBSERVATIONS	KNOWN CLUSTER	SUPPORT ORIGINS	DISPOSITION
09	**01**	**00**	**HOLD**
preserved	Origin A	for this broad claim	verify another relation

What changed in the bad summary

- Nine report observations became nine apparent origin paths.
- Apparent plurality became support for a broader claim.
- The unknown relationship between origin, stance, and claim disappeared.
- A fluent answer concealed the earlier accounting decision.

> **The correction is not “count less.”** *It is: count the right unit under a stated rule, preserve the reports, and keep unresolved lineage unresolved.*

NEXT PERMITTED STEP

Inspect the launch announcement; seek a separately authored benchmark or failure report; record its origin relation; then assess the exact claim it supports or refutes.

Preserve observations without inventing support.

This is a static illustration, not a dataset, provenance audit, or runtime. Rows are unordered and all nine report records remain visible.

RECORD	KIND	DECLARED RELATION	ACCOUNTING
O01	Report observation	Origin A · **DEPENDENT**	Preserve; not separately rooted support under this rule.
O02	Report observation	Origin A · **DEPENDENT**	Preserve; not separately rooted support under this rule.
O03	Report observation	Origin A · **DEPENDENT**	Preserve; not separately rooted support under this rule.
O04	Report observation	Origin A · **DEPENDENT**	Preserve; not separately rooted support under this rule.
O05	Report observation	Origin A · **DEPENDENT**	Preserve; not separately rooted support under this rule.
O06	Report observation	Origin A · **DEPENDENT**	Preserve; not separately rooted support under this rule.
O07	Report observation	Origin A · **DEPENDENT**	Preserve; not separately rooted support under this rule.
O08	Report observation	Origin A · **DEPENDENT**	Preserve; not separately rooted support under this rule.
O09	Report observation	Origin A · **DEPENDENT**	Preserve; not separately rooted support under this rule.

RELATION STATES

DPND · traceable to another observed path
INDP · separate origin by benchmark stipulation only
UNKN · unresolved; never guess
NONE · no relation value supplied

CLAIM STATE

INSUFFICIENT

Origin A exists, but its stance toward “broadly validated” has not been established. B1 and C1 are contrast roots with support unassessed.

HUMAN DISPOSITION

HOLD · VERIFY ANOTHER ORIGIN RELATION

No automatic admission, rejection, truth verdict, or provenance discovery.

03 · DISTINCTION CONTRACT

Keep unlike judgments unlike.

The point is error visibility, not one schema per concept. When distinctions collapse, the system loses the ability to explain where a bad route began.

KEEP SEPARATE	DO NOT SUBSTITUTE	WHY
Source identity	Artifact identity	One source can publish many artifacts; one artifact can move.
Provenance	Correctness	A false claim can have perfect lineage.
Recurrence	Corroboration	Copies can recur without a new root.
Origin relation	Claim support	A separate root can be irrelevant; a copy can contain a useful span.
Source authority	Universal trust	Authority is claim-, role-, domain-, and time-scoped.
Claim support	Citation presence	A citation can be irrelevant, contradictory, or too broad.
Relevance	General importance	Relevance belongs to the current decision.
Technical access	Authorization	Reachability is not permission to use or retain.
Enrichment value	Action priority	More learning can help without being the next permitted step.
Action priority	Truth	A sandbox step is a decision, not a factual verdict.
Owner disposition	External fact	Accept, defer, and override are accountable choices.
Outcome	Retroactive truth	A later result updates policy; it does not rewrite history.

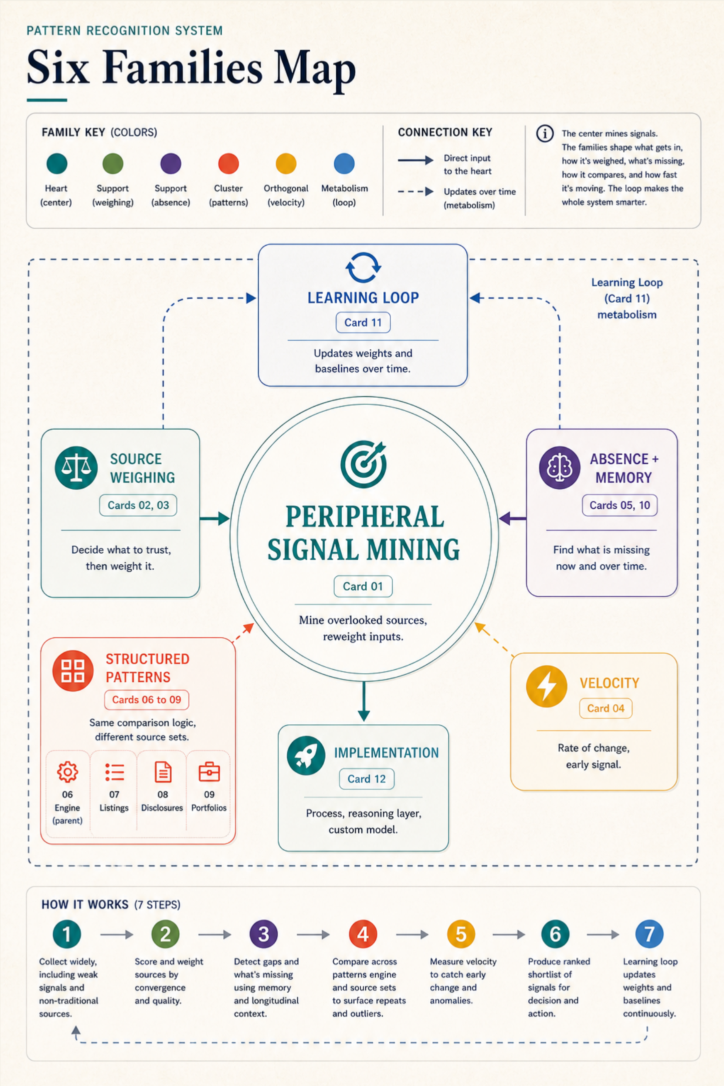
D04 · ORIGIN RELATION

Shared known origin · unresolved · separately rooted by stipulation

The historical `independence` label remains a compatibility alias. V15 does not equate this field with real-world causal, editorial, methodological, or epistemic independence.

Preserve v13; do not mistake it for current topology.

The recovered diagram is historical evidence of the concept's origin. V15 keeps it unchanged and renders the current architecture in live text and structured records.



ARTIFACT	STATUS	BOUNDARY
v13 diagram PNG	Exact hash verified	1024×1536; SHA-256 8a8204…3ae; immutable historical anchor
Rendered DOM snapshot	Reference capture	Useful rendered state; not the original standalone source
Standalone v13 HTML	Unavailable	Expected byte identity remains unverified

Six mechanism families; two loops; one preserved history.

The map is a reviewable decomposition, not a maturity model, a required service topology, or a proven minimum.

F1

Intent and authorization envelope

Define the question, consequence, permissions, baselines, and budgets before information is allowed to influence the workflow.

Components · C01

F2

Evidence spine

Acquire, identify, normalize, version, and trace artifacts without confusing capture with acceptance.

Components · C02, C03

F3

Relationship and claim graph

Represent grouping, common origin, recurrence, comparison, atomic claims, support, contradiction, and gaps.

Components · C04, C05

F4

Discrimination policy

Keep judgment dimensions separate, choose bounded next actions, and prepare an inspectable context packet.

Components · C06, C07, C08

F5

Human disposition and memory

Capture acceptance, rejection, deferral, override, and versioned retention without converting preference into fact.

Components · C09, C10

F6

Outcome feedback

Compare expected and observed outcomes and propose revisable policy updates.

Components · C11

FAST EVIDENCE LOOP	SLOW LEARNING LOOP
Brief → acquire → identify → relate → assess → route → package → owner disposition	Used packet → defined outcome → comparison → proposed update → human approval → new version
Returns to acquisition when a consequential gap warrants bounded search.	Never silently rewrites observations, prior decisions, or policy history.

From intent to origin-aware relationships

Every card names a responsibility, a failure watch, and its evidence maturity. The complete canonical specification remains in the HTML and machine-readable map.

C01 · F1

Decision brief and authorization envelope

A versioned declaration of the question, intended use, decision owner, stakes, expected baselines, allowed operations, sensitive-source rules, and resource limits.

Failure watch: An underspecified question makes relevance arbitrary.; Permissions are inferred from technical access.

Evidence status: Prior Art Supported Conceptual Synthesis

C02 · F2

Acquisition controller

A policy-governed mechanism that proposes, authorizes, records, and stops retrieval or collection actions.

Failure watch: Search scope expands without authorization.; Similarity or novelty is mistaken for value.

Evidence status: Prior Art Supported Design Hypothesis

C03 · F2

Source, artifact, normalization, and provenance spine

A set of stable identities and append-only derivation records for sources, artifacts, captures, versions, transformations, agents, and timestamps.

Failure watch: Two reports are merged because names look alike.; One artifact is duplicated and counted twice.

Evidence status: Prior Art Supported Conceptual Synthesis

C04 · F3

Relationship, recurrence, common-origin, and gap graph

A typed graph linking sources, artifacts, events, groups, observations, time points, copies, derivations, recurrences, comparisons, and expected-but-missing perspectives.

Failure watch: Syndicated copies are counted as distinct-origin support.; Unknown origin is silently classified as separately rooted.

Evidence status: Prior Art Supported Design Hypothesis

From exact claims to a bounded context packet

Every card names a responsibility, a failure watch, and its evidence maturity. The complete canonical specification remains in the HTML and machine-readable map.

C05 · F3

Claim, evidence, comparison, and contradiction graph

A claim-level representation connecting atomic propositions to exact evidence spans, support states, qualifications, contradictions, alternatives, and unresolved questions.

Failure watch: Claims remain too broad to test.; Citation presence is treated as support.

Evidence status: Prior Art Supported Conceptual Synthesis

C06 · F4

Multidimensional assessment

An inspectable set of separate, task-scoped judgments for attention priority, domain source authority, claim support, origin relation or stipulated distinctness, relevance, enrichment value, action priority, and owner disposition. Uncertainty and possible consequence are explicit qualifying attributes, not interchangeable scoring dimensions.

Failure watch: A universal trust score launders one dimension into another.; Owner interest becomes endorsement.

Evidence status: Conceptual Synthesis Requiring Evaluation

C07 · F4

Enrichment, stopping, and action router

A policy that compares next actions under uncertainty and resource constraints, including acquire, compare, enrich, clarify, answer, answer provisionally, hold, defer, escalate, or refuse.

Failure watch: An enrichment-value estimate becomes acceptance or dictates action priority.; A brittle utility estimate creates false precision.

Evidence status: Prior Art Supported Design Hypothesis

C08 · F4

Bounded context packet

A versioned package of selected content plus provenance, claim links, reasons for inclusion and exclusion, unresolved states, budgets, and generation constraints.

Failure watch: Compression changes a claim or drops a qualifier.; Ordering over-amplifies convenient evidence.

Evidence status: Design Hypothesis With Adjacent Precedent

From human correction to revisable learning

Every card names a responsibility, a failure watch, and its evidence maturity. The complete canonical specification remains in the HTML and machine-readable map.

C09 · F5

Owner disposition, review, and override

An explicit human control surface for accept, reject, defer, hold, override, request enrichment, correct relationships, and revise task constraints.

Failure watch: Rubber-stamping or automation bias.; The reviewer sees a conclusion without its evidence path.

Evidence status: Prior Art Supported Design Hypothesis

C10 · F5

Versioned evidence, decision, and memory ledger

Append-only retention of observations, derived interpretations, decisions, context packets, generated outputs, corrections, and supersession relationships.

Failure watch: A summary overwrites raw evidence.; Retrieved memory loses source or epistemic type.

Evidence status: Design Hypothesis With Adjacent Precedent

C11 · F6

Outcome feedback and revisable policy update

A controlled loop that compares recorded predictions, decisions, costs, and expected outcomes with defined later observations, then proposes bounded updates.

Failure watch: No outcome was defined before the decision.; A proxy is treated as the desired consequence.

Evidence status: Design And Empirical Hypothesis

MINIMUM IMPLEMENTATION CLAIM

Task and permission framing; identity and provenance; claim and relationship representation; separate assessment dimensions; bounded routing; human correction; versioned outcome feedback whenever learning is claimed.

09 · WORKED APPLICATION

What changes after the receipt?

The illustration is not another telling of the nine-report example. It shows how a corrected accounting record changes the next permitted action without deciding truth.

BEFORE	CORRECTION	AFTER
Nine appearances were summarized as nine supporting sources.	All nine are linked to Origin A; support for the broad claim remains unassessed.	The route changes from premature acceptance to HOLD plus one bounded verification step.
Unknowns disappeared in prose.	UNKN remains a first-class relation state.	The packet states what is missing and when to stop.

DECISION IMPLICATION

A sandbox pilot can be authorized without endorsing “broadly validated.” Permission, evidence status, and action priority remain different records.

E2 is a labeled AI-generated illustration. It is not a dataset, finding, independent validation, or required interpretive evidence.

The closest precedents remove the broad novelty claim.

V15 keeps the contribution narrow by distinguishing established mechanisms, direct comparators, adjacent methods, and the remaining supplied-cue experiment.

PRECEDENT	WHAT IT ESTABLISHES	WHAT IT BLOCKS
Dong et al. · VLDB 2009	Copying-aware truth discovery and source dependence.	No novelty claim for dependence-aware aggregation.
Zhang, Ives & Roth · ACL 2020	Natural-language claim-provenance graphs with inferred source/statement relations.	Closest direct precedent; supplied oracle cues must be distinguished from provenance inference.
Senn · BMC MRM 2009	Shared studies or analyses can be double-counted.	No novelty claim for choosing an effective evidence unit.
Cochrane Handbook · current	Multiple reports must be collated around the underlying study.	No novelty claim for preserving reports while avoiding study double count.
Greenberg · BMJ 2009	Citation practices can amplify an unsupported claim into apparent authority.	Citation count is not independent corroboration.
NEWS-COPY · arXiv 2022	Same-original duplicates under abridgement and OCR noise.	No novelty claim for noisy duplicate grouping.
Newswire · NeurIPS 2024	Historical reproduction clusters separate appearances from reproduced articles.	Outlet recurrence cannot be equated with perspectives or truth.
MMR / SetR / NEST / RARE / Schelpe	Diversity-aware, set-wise, redundancy-sensitive retrieval, and byte-exact deduplication precedents.	No novelty claim for redundancy penalties, joint set selection, or exact duplicate removal.
Strittmatter et al. · 2024	Human judgments of evidential dependence in fictitious scenarios.	Human sensitivity and effective-independence questions have direct precedent.

NOVELTY DISCIPLINE

The project does not introduce copying detection, deduplication, claim graphs, diversity retrieval, conflict handling, or evidence-unit accounting.

What remains after RAG and provenance work are credited?

Several current manuscripts are useful comparisons, but publication status, task mismatch, and origin-ground-truth limits stay visible.

WORK	DIRECT VALUE	BOUNDARY
RAMDocs	Conflicting-evidence robustness	Adjacent; not origin accounting
Li, Padman & Krishnan	Cross-institution source-set answer variation	Unreviewed manuscript; source answers are not derivation
EvidentialRAG	Conflict and uncertainty fusion	Unreviewed manuscript; not oracle-origin cue isolation
Naphade	Distinct vs paraphrased opposing evidence	Unreviewed; distinct documents are not verified origins
Ross et al.	Exact duplicate, paraphrase, and diverse retrieval sets	Unreviewed; different estimand and no real-world origin proof

RESIDUAL CONCEPTUAL CONTRIBUTION

A boundary-preserving synthesis that makes identity, origin relation, claim support, permission, cost, routing, and human disposition jointly inspectable before generation.

RESIDUAL EMPIRICAL QUESTION

Does a supplied typed origin cue change origin counting beyond an explicit rule when evidence, prompt budget, model, and output contract are held fixed?

This is not provenance discovery, a new RAG architecture, or broad mechanism novelty.

The strongest objections are design requirements.

A defensible framework states how it can fail, when simpler methods should win, and when the name or decomposition should be retired.

Old work under a new label

Credit mature provenance, evidence-synthesis, retrieval, HCI, and learning literatures; keep only synthesis and narrow tests.

A gatekeeper in quality-control clothing

Expose exclusions, unknowns, permissions, appeal, and source coverage; let correction change the packet.

Provenance as rigor theater

Require each receipt field to change a consequential route or remove it.

More costly than the error

Compare against strong simple retrieval-plus-citation baselines under matched cost.

Decorative human review

Expose exact spans, relations, omissions, and reasons; measure whether people can correct them.

Feedback optimizes the wrong proxy

Predefine target, horizon, exposure, confounders, and approval; never silently update.

The name does harm

Test reader comprehension and prefer Context Judgment Layer if the technical definition fails.

RETIRE FOR A TASK WHEN...

A simpler baseline performs equivalently at lower cost; the distinctions are not usable; origin grouping suppresses valid convergence; review and permissions become ceremonial; feedback amplifies contaminated proxies; or the name continues to misstate the thesis.

Does an oracle origin cue add anything beyond an explicit rule?

One frozen model, one fixed synthetic corpus, one primary paired contrast. The model, tokenizer, prompts, manifests, and run environment must be locked before any primary output.

CONDITION	RULE	RELATION SLOTS	ROLE
F0	Ordinary bounded evidence assessment	NONE	Secondary baseline
F1	Count distinct pathways; do not count repeated/derived reports; preserve unknown	NONE	Rule-only comparator
F2	Byte-identical F1 instruction	DPND / INDP / UNKN from stipulated graph	Oracle-cue condition

PRIMARY ESTIMAND

F2 – F1 on conservative false-corroboration risk across all 300 assigned primary bundles.

F0 is descriptive. F3 does not exist. T1 is not a condition. Invalid outputs remain risk events rather than disappearing from the denominator.

PARITY LOCK

F1/F2 must preserve exact report bytes and order, exact input-byte length, exact system and instruction bytes, and exact token count under the selected frozen tokenizer. The local regex tokenizer is only a development surrogate.

Fixed denominators make failure visible.

The generator is deterministic, fictional, offline, and provenance-controlled. Primary and safety sets are frozen before output; parse failures never shrink them.

SPLIT	TOTAL	ONE ORIGIN	MULTIPLE	UNKNOWN	CONFLICT
dev	80	20	20	20	20
pilot	40	10	10	10	10
primary	300	75	75	75	75
stress	60	15	15	15	15

PRIMARY · FC_cons

False corroboration

Risk = invalid output, or a valid count of at least two supporting origins when the manifest certifies none/single or withholds certification as unknown.

Denominator: all 300 assigned primary bundles.

SAFETY · VOR

Valid-origin recall

On the frozen multiple-origin subset, a valid output must count at least two supporting origins and cite evidence spanning at least two stipulated support roots.

Margin: paired lower bound must exceed −0.05.

UNKNOWN-ORIGIN TRUTH BOUNDARY

Every prompt-visible value is UNKN. A valid assertion of two or more paths is conservatively risk-coded because the relation is uncertified—not because the latent world is known to contain zero or one origin.

A favorable number is insufficient without a valid route to it.

The confirmatory decision has one superiority test and one safety gate. Everything else is descriptive or diagnostic.

PRIMARY DECISION	SAFETY DECISION
Beneficial F2−F1 paired risk difference; exact two-sided McNemar/binomial $p<.05$; 95% paired bootstrap interval upper bound below 0; N=300.	On fixed M, report F1/F2 VOR, delta, membership hash, and prespecified one-sided 95% lower bound; pass only above −0.05.
Report whether the −0.08 planning benchmark was reached; it is not another hidden test.	Interval method and coverage must be frozen and simulated at actual |M| before preregistration.

Fail closed before any empirical run

- Corpus integrity: exact hashes, split-family isolation, graph invariants, and all-UNKN prompt invariant.
- Prompt parity: exact bytes, order, instruction fields, and intended-tokenizer equality.
- Leakage: blocked surface classifier with frozen ceiling and interval; semantic audit with adjudication.
- Execution receipt: exact model/checkpoint/tokenizer/runtime/hardware/decoding plus immutable raw bytes and hashes.
- Analysis lock: A=300, fixed M, 10,000-resample primary interval, declared safety interval, coverage simulations.

CURRENT STOP LINE

No model or tokenizer is selected; no pilot or primary output exists; no preregistration, participant, provider, deployment, or publication is authorized. Offline readiness does not cross that line.

T1 is descriptive; an unfavorable result is publishable evidence.

Natural syndication can stress parts of the accounting problem, but available public data do not supply the ground truth required by the confirmatory estimand.

SOURCE	BOUNDED USE	BLOCKING LIMIT
NEWS-COPY	Same-original / dependent fixtures	Nonduplicates are not verified independent origins; rights unresolved
Newswire	Reproduction clusters and recurrence context	Cluster rows lack claim stance, evidence spans, support/refute origins, and multiple-origin truth

T1 BOUNDARY

Separate rights and annotation manifest · descriptive only · outside A, M, primary/safety denominators, confidence intervals, tests, and effect estimates · never F3

Locked interpretations

OBSERVED PATTERN	REQUIRED INTERPRETATION
F1 and F2 beat F0; F2 ties F1	Credit the explicit rule, not typed metadata.
F2 reduces FC but harms VOR	Reject the cue as unsafe; it suppresses valid stipulated convergence.
Effect survives only superficial metadata	Report a shortcut, formatting, or direct-code result.
Null, negative, harmful, or unstable	Preserve and report it; do not change the endpoint, denominator, or run until favorable.

Sixteen constraints that cannot be smoothed away

The package is complete only if the absences are as legible as the proposal.

01 No empirical evaluation	**02** No broad novelty finding	**03** No proven minimum decomposition	**04** No validated constructs
05 No provenance discovery	**06** No real-world independence claim	**07** No truth result	**08** Open-world evidence remains incomplete
09 Costs are unknown	**10** Human control is unproven	**11** Memory can amplify error	**12** Transfer remains unresolved and rights-gated
13 Historical standalone v13 HTML is unavailable	**14** Product cases are circular if treated as evidence	**15** The name may fail reader comprehension	**16** No publication or deployment authorization

A COMPLETE CONCEPTUAL SITE CAN PRECEDE RESULTS

The empirical paper and any result surface cannot. Until a separately authorized run passes every gate, the Lab remains a protocol and all result language remains absent.

Keep the question visible enough to answer honestly.

An AI system will make context judgments whether or not it names them. The proposal is to expose those judgments early enough to contest—then retire the machinery wherever it does not earn its cost.

> *The important move is not to count less. It is to count the right unit under a stated rule and preserve what remains unknown.*

Owner questions

- Does the nine-report receipt make the thesis concrete without overclaiming?
- Is “Discrimination Layer” worth retaining, or should the public name become “Context Judgment Layer”?
- Are the six-family / eleven-component map and its compatibility IDs ready to freeze for owner review?
- Should the narrow F2−F1 program remain the first paper, contingent on a later exact model/tokenizer/run authorization?
- Should T1 remain deferred until rights and field-level annotation feasibility are independently cleared?

CANONICAL READING SURFACE

site/

Semantic local HTML with Essay, Explore, and Lab tracks.

source/THOUGHT_PIECE_V15.md

EXECUTION SPECIFICATION

research/

Prospectus, protocol, readiness memo, schemas, offline tools, and review receipts.

No external run is authorized by these files.

FINAL STATUS

LOCAL OWNER REVIEW · CONCEPTUAL SYNTHESIS · UNRUN RESEARCH PROGRAM · NOT PUBLISHED